

TAMIL NADU STATE BOARD - STANDARD 12 MATHEMATICS VOLUME 1

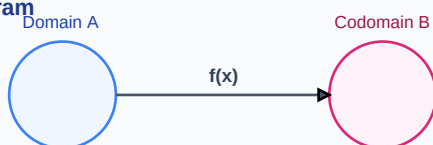
CHAPTER 1: APPLICATIONS OF MATRICES AND DETERMINANTS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Mathematics Volume 1
Chapter Name:	Applications of Matrices and Determinants

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Singular Matrix	A square matrix A is singular if its determinant is zero, meaning $ A = 0$. It does not possess a multiplicative inverse.
Adjoint Matrix	The transpose of the cofactor matrix of a given square matrix A , denoted by $\text{adj}(A)$.
Rank of Matrix	The maximum number of linearly independent row or column vectors in matrix A , determined by reducing to row-echelon form.

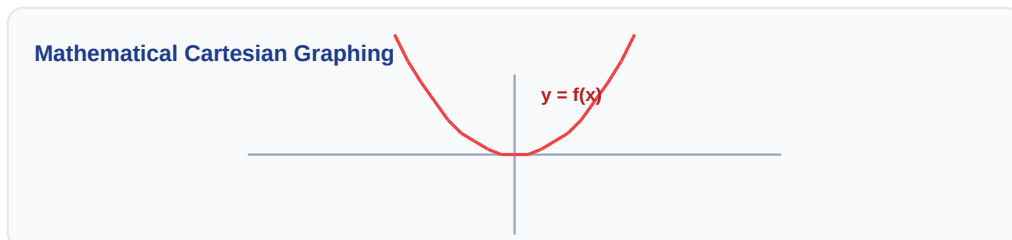
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Inverse Formula: $A^{-1} = (1 / |A|) * \text{adj}(A)$
- Product Property: $\text{adj}(AB) = \text{adj}(B) * \text{adj}(A)$
- System Consistency (Rouche-Capelli): Consistent if $\text{rank}(A) = \text{rank}([A|B])$
- Cramer's Rule: $x_i = \Delta_i / \Delta$ for $\Delta \neq 0$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find the inverse of matrix $A = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$.

Step-by-step Solution:

Determinant $|A| = 2(-3) - (-1)(5) = -6 + 5 = -1$ (non-zero).

Cofactor matrix: $C_{11}=-3$, $C_{12}=-5$, $C_{21}=1$, $C_{22}=2$.

Adjoint $\text{adj}(A) = \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}$.

Inverse $A^{-1} = (1/-1) * \text{adj}(A) = \begin{bmatrix} 3 & -1 \\ 5 & -2 \end{bmatrix}$.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Investigate the consistency of the system: $x + y = 3$, $2x + 2y = 6$.

Analytical Solution Strategy:

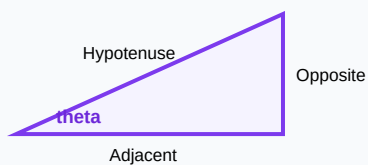
Coefficient matrix $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $\text{rank}(A) = 1$.

Augmented matrix $[A|B] = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 2 & 6 \end{bmatrix}$.

Row 2 $\rightarrow$ Row 2 - 2 * Row 1 $\Rightarrow \begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$.

Rank of augmented matrix is 1. Since $\text{rank}(A) = \text{rank}([A|B]) = 1 < 2$ (variables), the system is consistent with infinitely many solutions.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State when a system of linear equations is inconsistent.

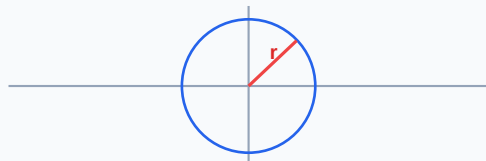
Q2 (2-Mark): If A is a square matrix of order 3 and $|A| = 5$, find $|\text{adj}(A)|$.

Q3 (2-Mark): Solve using Cramer's rule: $2x - y = 5$, $x + 3y = -1$.

Q4 (5-Mark): Reduce the matrix $\begin{bmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \end{bmatrix}$ to row-echelon form and find its rank.

Q5 (5-Mark): Prove that $A * \text{adj}(A) = |A| * I$.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Cryptography

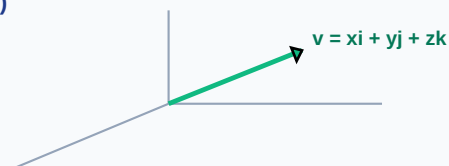
Matrix multiplication and inversion encode and decode secure communications via Hill Cipher models.

Application Case: Graphics Engineering

Linear transformation matrices map coordinate transformations (scale, rotate, translate) in rendering engines.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

